SIMATS ENGINEERING
ASSESSMENT TOOL 3

COURSE CODE: MLA 02 **COURSE NAME: Fundamentals Of Machine Learning**

COURSE OUTCOME COVERED

CO5: *Create graphical models to analyze and infer patterns in complex probabilistic systems for structured decision-making.(BL5)*

Assessment Tool Weightage: 10%

QUESTIONS: 10 **Total Marks:50**

Annexure A

Mock Interview

Code of Conduct:

I ____________________(Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student:

Annexure A

Mock Interview

Q1. “Suppose you are working on an industrial monitoring system. How would you use Bayesian Networks to predict machine failures?” (5 Marks – 8 Mins)

Q2. “Imagine you are asked in an interview to design an image denoising solution. How would Markov Random Fields help model neighborhood pixel dependencies?” (5 Marks – 8 Mins)

Q3. “If a company wants to build a speech recognition system, how would you apply a Hidden Markov Model to process sequential audio signals?” (5 Marks – 8 Mins)

Q4. “During a technical interview, how would you explain the use of Conditional Random Fields for sequence labeling tasks such as part-of-speech tagging?” (5 Marks – 8 Mins)

Q5. “Suppose an interviewer asks how conditional independence simplifies probabilistic inference. How would you justify it with a graphical model example?” (5 Marks – 8 Mins)

Q6. “In a healthcare diagnosis problem, how would you compare directed graphical models and undirected graphical models for representing symptom relationships?” (5 Marks – 8 Mins)

Q7. “If patient symptom data is partially missing, how would you perform Bayesian inference to still make an accurate diagnosis?” (5 Marks – 8 Mins)

Q8. “How would you use a Hidden Markov Model to analyze customer purchase sequences in an e-commerce recommendation system?” (5 Marks – 8 Mins)

Q9. “In an NLP interview task, how would you explain why Conditional Random Fields often outperform Hidden Markov Models in named entity recognition?” (5 Marks – 8 Mins)

Q10. “Suppose you are asked to improve fraud detection. How would graphical model learning and generalization enhance prediction accuracy?” (5 Marks – 8 Mins)

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MOCK INTERVIEW

5. Conditional Independence

Conditional independence means that two variables becomes Independant when we know the value of another variable is wet. some information about rain, wet road, and Traffic. If we already know whether the road is wet, some information about rain may not provide additional information about traffic. Graphical models we conditional independence to reduce the no.of probabilities that need to be calculated. This make probilistic inference faster and simpler.

6. Directed vs Undirected Models:

In. health care diagnosis, a directed graphical model, such as a Bayesian Network, represents relationship using directed arrows. Bayesian Network are example of directed models. while Markov random Fields are example of undirected m

7. Missing Patient Symptoms.

If some symptoms are missing, Bayesian Inference can still help us make a diagonosis. we use the symptoms that are available and calculate the probability of different dieases. Then we select the disease with the high probability. So, we can make a prediction even with incomplete data.

10. Graphical Models - Fraud Detection:

Graphical model can learn relationship between transaction features such as amount, location, time and user behavior. The model learns pattern from previous transactions. It can then use these patterns to identify new suspicious transactions. Good learning and generalization help improve fraud detection accuracy.

RUBRICS FOR CALCULATION

Criteria	Weightage	Level 4 (Exemplary)	Level 3 (Proficient)	Level 2 (Developing)	Level 1 (Beginning)
Technical Knowledge	25	Demonstrates strong and accurate subject knowledge with in-depth understanding	Shows good knowledge with minor gaps	Basic knowledge with noticeable errors	Lacks fundamental technical knowledge
Problem Solving	25	Solves problems logically and applies concepts effectively in real scenarios	Applies concepts with moderate accuracy	Limited problem-solving ability; requires guidance	Unable to solve or apply concepts
Analytical Thinking	15	Analyzes questions critically and provides well-structured, logical answers	Provides reasonable analysis with some structure	Superficial or partially structured responses	No analytical thinking demonstrated
Communication	15	Communicates clearly, confidently, and professionally with proper terminology	Communicates well with minor hesitation	Communication lacks clarity or confidence	Unable to communicate effectively
Confidence	15	Displays high confidence, positive attitude, and professional	Shows good confidence with minor issues	Low confidence; inconsistent behavior	Lacks confidence and professionalism

		behavior throughout			
Response to Questions	10	Responds accurately, promptly, and elaborates with relevant examples	Responds correctly but with limited depth	Partial responses; needs prompting	Unable to respond appropriatel y